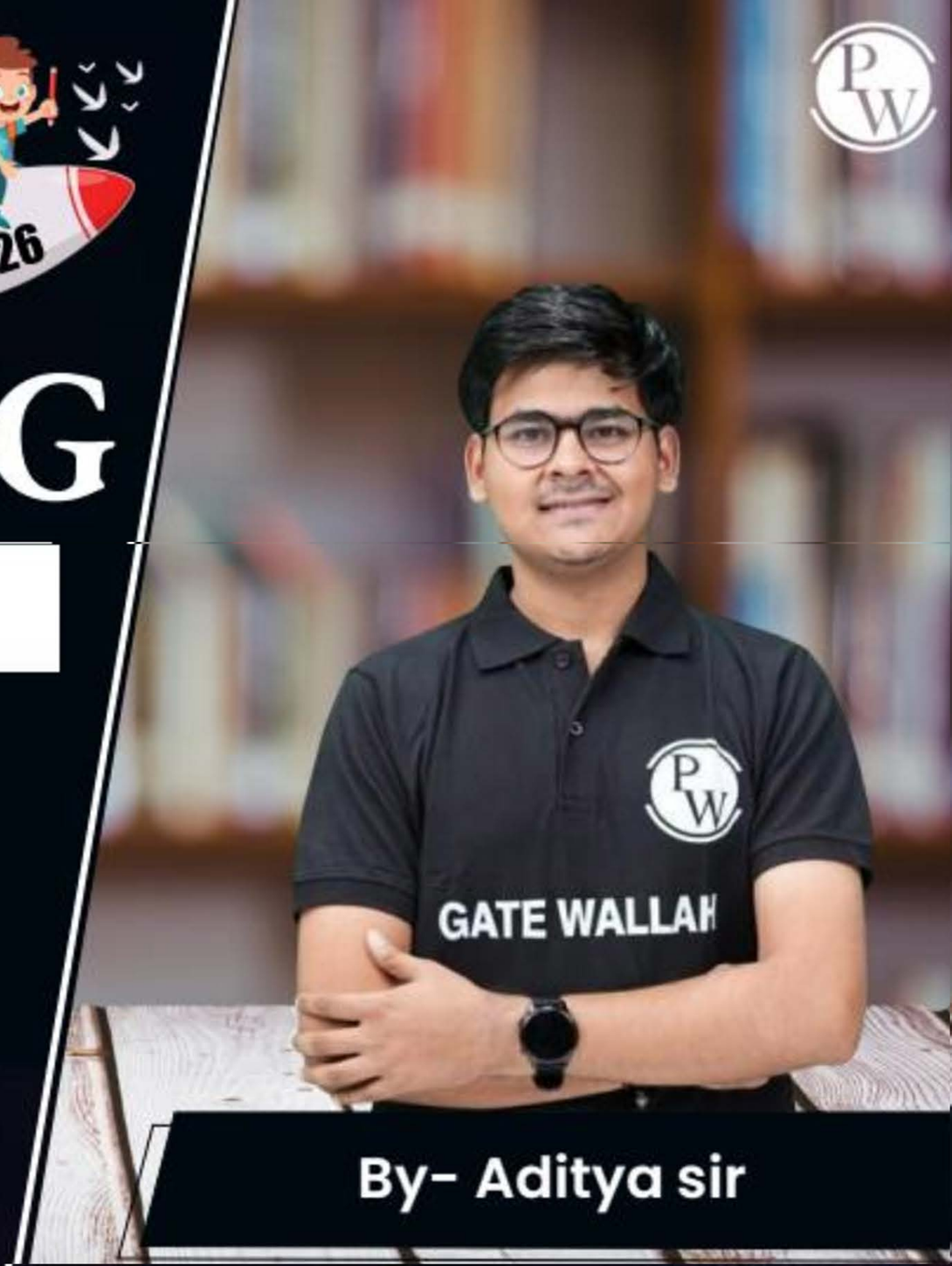
CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 02



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Intro to Greedy Algo

→ Framework

→ 1) Fractional Knapsack

Topics to be Covered



- Topic
- Topic
- Topic

Fractional knapsack

→ Imp Terminologies

→ 2) Job Sequencing with deadline.



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: n = 7, M = 15

int(ans)

Object	Profit	Weight
01	10	2
02	5	3
03	15	5
04	7	7
05	6	1
06	18	4
07	3	1

P/w

10/2 = 5

5/3 = 1.66

15/5 = 3

7/7 = 1

6/1 = 6

18/4 = 4.5

3/1 = 3

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:
Given: n = 7, M = 15

Object	Profit	Weight
01	10	2
02	5	3
03	15	5
04	7	7
05	6	1
06	18	4
07	3	1

int(ans)

P/w

1 → 10/2 = 5 — 2
2 → 5/3 = 1.66 — 6
3 → 15/5 = 3 — 5
4 → 7/7 = 1 — 7
5 → 6/1 = 6 — 1
6 → 18/4 = 4.5 — 3
7 → 3/1 = 3 — 4

Xi

01 → $X_1 = 1$
02 → $X_2 = 2/3$
03 → $X_3 = 1$
04 → $X_4 = 0$
05 → $X_5 = 1$
06 → $X_6 = 1$
07 → $X_7 = 1$

$M = 15 - 1$
$= 14 - 2$
$= 12 - 4$
$= 8 - 1$
$= 7 - 5$
$= 2 - 2 \Rightarrow 0$

Total Profit $= \sum P_i x_i$

$= 10 + 15 + 6 + 18 + 3 + 5 \times 2/3 + 7 \times 0$

$= 25 + 27 + 3.33$

$= 52 + 3.33$

$= \underline{\underline{55.33}}$



Topic : Greedy Algorithm

Fractional – Knapsack

```
Procedure GREDY_KNAPSACK(P, W, M, X, n)
// P(1 : n) and W(1 : n) contain the profites and weights respectively of the n//
// knapsack size and X(1 : n) is the solution vector//
Real P(1 : m), W(1 : n), X (1 : n), M, cu;
integer I, n;
x ← 0 // initialize solution to zero//
cu ← M // cu = remaining knapsack capacity//
for i ← 1 to n do
    if W(i) > cu then exit endif
    X(i) ← 1
```

Imp:- { Objects are arranged as per decreasing order of P/w values }



Topic : Greedy Algorithm

```
        cu ← cu –W(i)
Repeat
    if i ≤ n then X(i) ← (cu/W(i))endif
end GREEDY–KNAPSACK
```

Fraction

[NAT]

(V. V. Imp) PYQ

#Q. Consider the weights and values of items listed below. Note that there is only one unit of each item.

By default

Profit

item number	Weight (in kg)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

[NAT]

#Q. The task is to pick a subset of these items such that their total weight is no more than 11 kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered lits. The total value of items picked by the greedy algorithm is denoted by V_{greedy}. The value of $V_{opt} - V_{greedy}$ is ____.

[NAT]

#Q. Consider the weights and values of items listed below. Note that there is only one unit of each item.

item number	Weight (in kg)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

P/w
60/10 = 6
28/7 = 4
20/4 = 5
24/2 = 12

(2) → Skip $x_1 = 0$
(4) → $x_2 = 1$
(3) → $x_3 = 1$
(1) $x_4 = 1$

$n = 11$

$V_{opt} = 60$

$n = 11 - 2$
$= 9 - 4$
$= 5$

$V_{greedy} = 24 + 20$
$= 44$

Overall avg $= 60 - 44$
$= 16$

Imp Terminologies



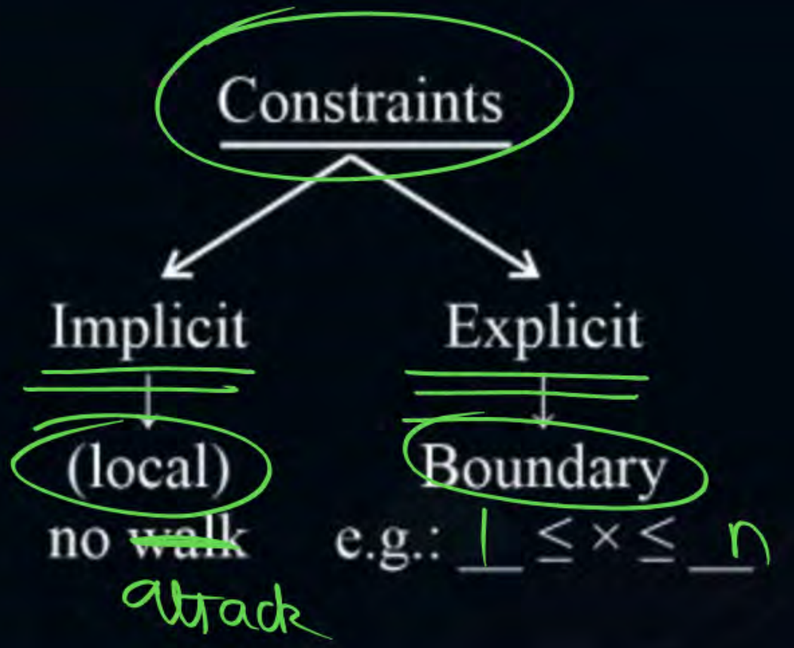
Topic : Greedy Algorithm

Important Terminologies:

1) Problem Definition:

2) Solution Space:

- All possible arrangements
- Only satisfies explicit constraints.



3) ~~Problem Definition:~~ Feasable Soln:-

- Those solutions from solution space that also satisfies implicit constraints.



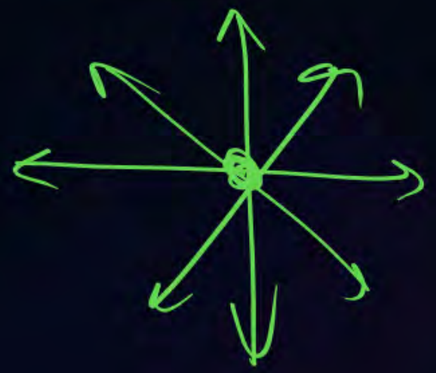
Topic : Greedy Algorithm

Example:

N-Queens Problem:

- Given n-Queens on a n × n board, arrange them such that no two queens attack each other.

E.g.: n = 4: q_1 q_2 q_3 q_4



	1	2	3	4
q_1				
q_2				
q_3				
q_4				



Topic : Greedy Algorithm

E.g.: n = 4: q_1 q_2 q_3 q_4

X[1 ... n]

X[i] → position of q_i in the ith row

$\{1 \le X_i \le n\}$

	1	2	3	4
q_1				
q_2				
q_3				
q_4				



Topic : Greedy Algorithm

E.g.: n = 4

X = [1, 2, 3, 4]

X = [2, 3, 1, 4]

X = [4, 3, 2, 1]

:

:

:

Size of solution space = n!

	1	2	3	4
q_1		•		
q_2			•	
q_3	•			
q_4				•

2, 3, 1, 4 ?

↳ not a Feasible Soln.

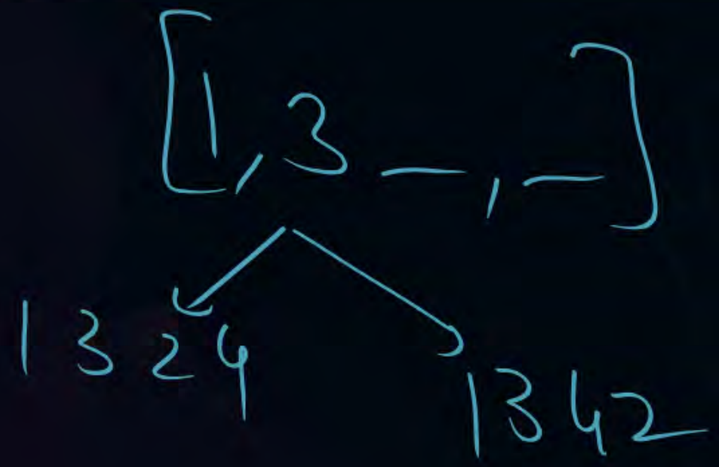


Topic : Greedy Algorithm

X = [1, 3, _, _]

- This is in solution space,
- But not feasible solution.

	1	2	3	4
q_1	q_1			
q_2			q_2	
q_3				
q_4				





Topic : Greedy Algorithm

X = [2, 4, 1, 3]

	1	2	3	4
q_1		q_1		
q_2				q_2
q_3	q_3			
q_4			q_4	

X = [3, 1, 4, 2]

	1	2	3	4
q_1			q_1	
q_2	q_2			
q_3				q_3
q_4		q_4		

- Both are feasible solution.



Topic : Greedy Algorithm

Important Terminologies:

4) Objective function:

- Optional
- Tries to optimize (minimize/maximize) a given criteria of the problem.



Topic : Greedy Algorithm

Example:

1) Knapsack → Maximize profit

2) Shortest Path → Minimize path cost



Topic : Greedy Algorithm

Important Terminologies:

5) Optimal Solution:

- It is that feasible solution that satisfies the objective function.
- Optimal solution always refers to the value and hence it is unique.

V. Imp

12

ABC
ADC



Topic : Greedy Algorithm

Example:



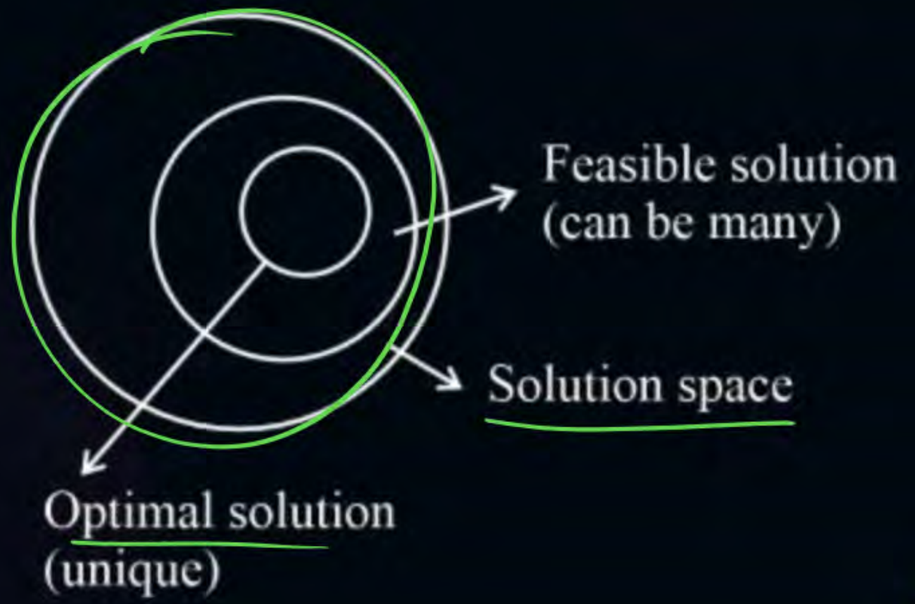
- Min cost A → C : 12 (unique)
- Shortest Path A → C :

  A → B → C : 5 + 7 = 12

  A → D → C : 8 + 4 = 12

[NAT]

#Q. What is the objective function of n-Queens problem?



- It does not have any objective function. Hence no optimal solution.
- It does have multiple feasible solutions.



Topic : Greedy Algorithm



Problem (P)

Decision Problem

- result is always either True/False.
- Feasible Solution(s)

e.g.: Searching, Sorting, n-Queens etc.



Optimization Problem

- Requires to determine a min/max value (optimal sol).
- Optimal solution
- Objective function

e.g.: Knapsack → maximizing profit
Shortest Path Problem → minimizing cost

2) Job - Sequencing with deadline

↳ OS

↳ [JSDJ]

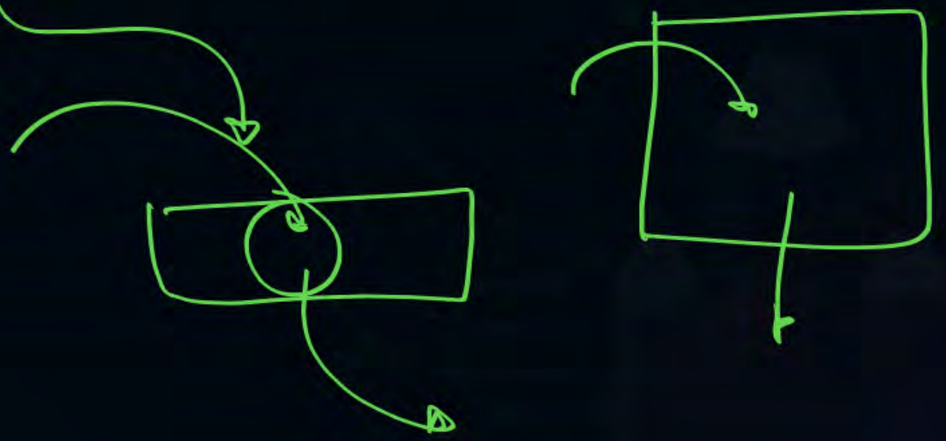


Topic : Greedy Algorithm

Problem 2:

Job Sequencing with Deadline (JSD):

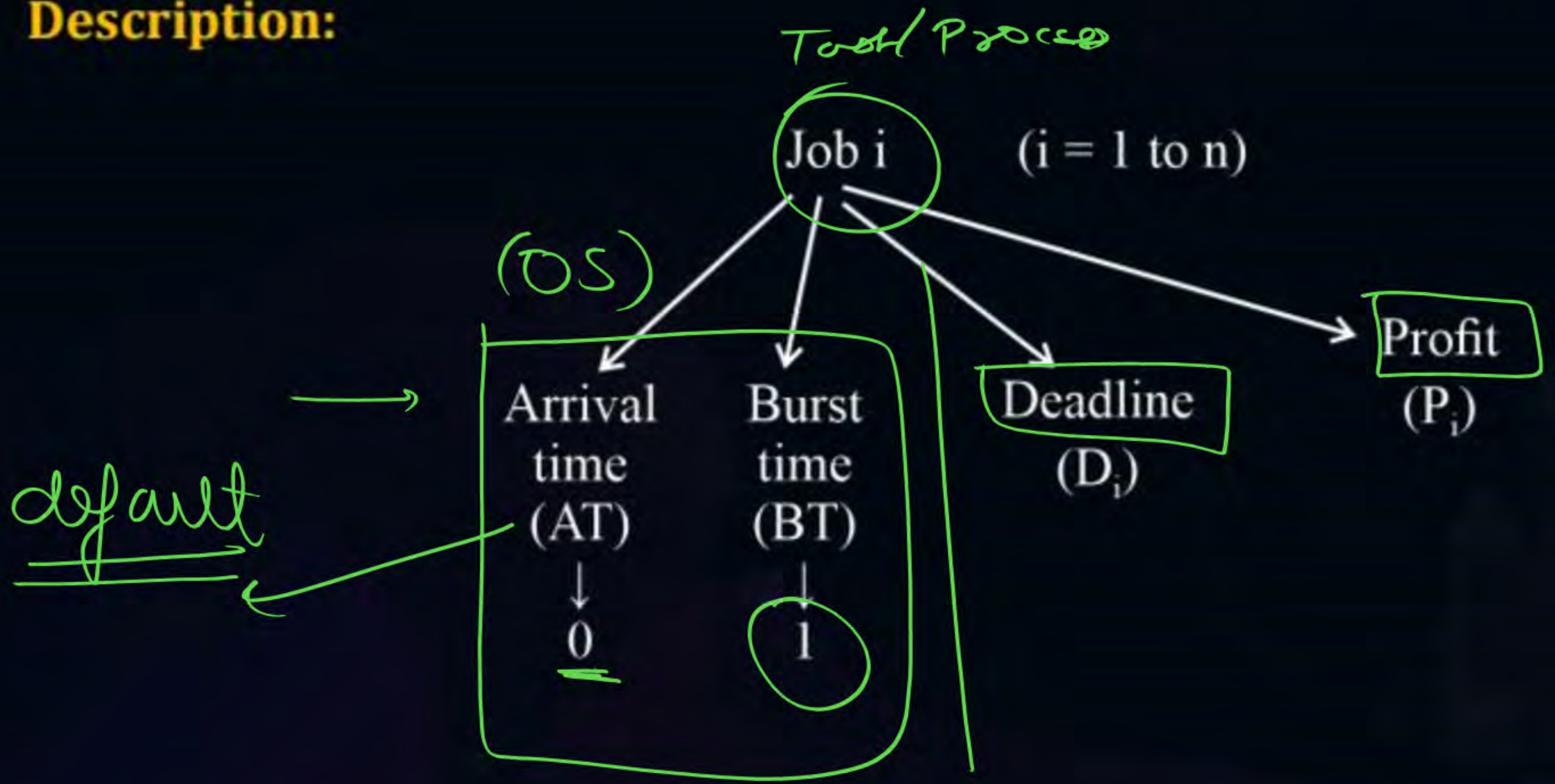
- Given a single CPU, Non-Preemptive Scheduling.
- Given n-jobs (Tasks/Processes)





Topic : Greedy Algorithm

Description:





Topic : Greedy Algorithm

Problem Statement:

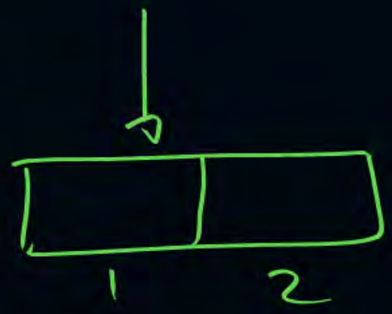
- Select a subset of 'n' given jobs, such that all the jobs in the subset are completed within their corresponding deadlines and generate the maximum profit.

Note: Only when the job gets completed with the given deadline, then only you get the associated profit (else not)



Topic : Greedy Algorithm

Soln = 280



Example:

$n = 4, J = \phi, |J| = 0$

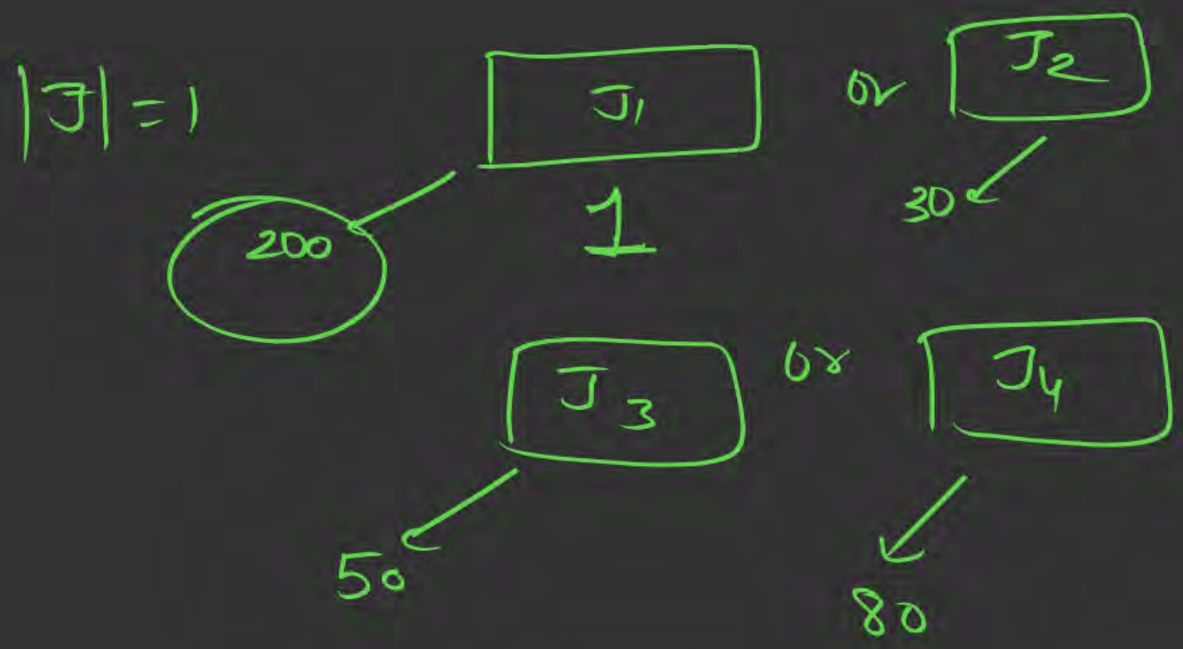
	Profit	Deadline
J1	200	2
J2	30	1
J3	50	2
J4	80	1

Soln:

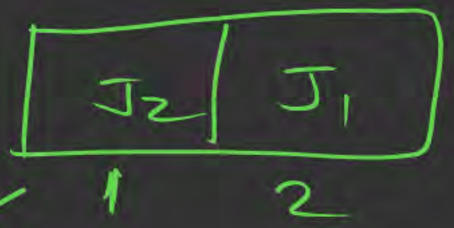
$|J| = 0$

Profit = 0

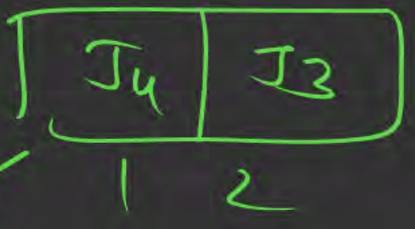



|J| = 1
J1
or
J2
200
1
30
J3
or
J4
50
80

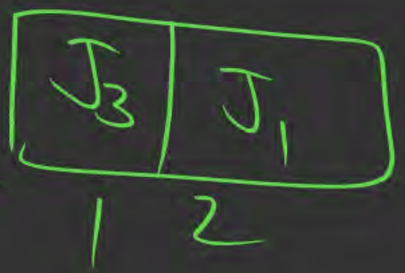
$|J| = 2$



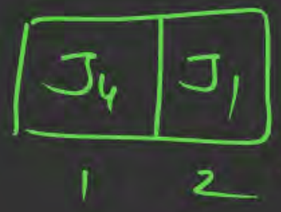
200+30
= 230



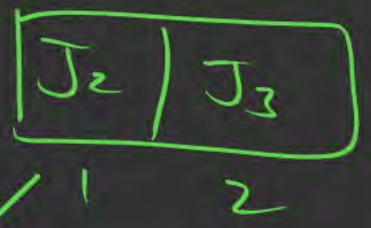
50+80
= 130



50+200
= 250



80+200
= 280



30+50=80



Topic : Greedy Algorithm

- This approach is Brute Force (Enumeration)

  1 Jobs → Size of solution space → $O(2^n)$

  Hence, not a good idea to solve like this.

- We need a Greedy Algorithm based approach.



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Knap ✓

Termino ✓

JSD



THANK - YOU